

CM0832 - MT5001 Elements of Set Theory

1. Sets

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Preliminaries

Textbook

Hrbacek and Jech ([1978] 1999). Introduction to Set Theory.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Acronyms

ZFC Zermelo-Fraenkel set theory with Choice

FOL First-Order Logic

First-Order Language of ZFC

Description

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The membership relation is denoted by $\in$.

The **non-logical symbols**, that is, the **language** $\mathcal{L}$ of the first-order theory **ZFC** is defined by $\mathcal{L} = \{\in\}$.

Properties

Definition

A **proposition** (or **statement**) is a sentence that can be assigned a truth value of **true** or **false**.

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The **domain** of a propositional function is the **domain of the discourse**.

The **codomain** of a propositional function is $\{\text{true}, \text{false}\}$.

Properties

Description

All the **properties (propositional functions)** of ZFC are stated using the membership relation and the logical constants (identity, unary and binary logical connectives and quantifiers).

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Example

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Since that we can prove that there is a **unique** set X with the property $\mathbf{P}(X)$ we can **add** a new **constant** symbol: $\emptyset$ (the empty set).

Properties

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We define the following binary property:

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Using the above property we can **add** a new binary **relation** symbol:

$X \subseteq Y$ (X is a subset of Y) $:= \mathbf{P}(X, Y)$.

Properties

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$Q(X, Y, Z)$: For every U , $U \in Z$ if and only if $U \in X$ and $U \in Y$.

Since that we can prove that for every X and Y there is a **unique** Z such that $Q(X, Y, Z)$, we can **add** a new binary **function** symbol:

$X \cap Y$ (the intersection of X and Y) $:=$ The unique Z such that $Q(X, Y, Z)$.

The Axioms

The axiom of existence

There exists a set which has no elements.

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The axiom of existence postulate the our universe of discourse is not void. This axiom postulates the existence of the empty set.

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The axiom of existence postulate the our universe of discourse is not void. This axiom postulates the existence of the empty set.[†]

[†]Strictly speaking, the universe of the discourse of FOL is not empty. For example, let $\mathbf{P}(x)$ be a property, then $(\forall x)\mathbf{P}(x) \rightarrow (\exists x)\mathbf{P}(x)$ is a theorem.

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Remark

The axiom of extensionality postulates that if two sets have the same elements, then they are identical.

The Axioms

Lemma (1.)3.1

There exists only one set with no elements.

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Proof

On the whiteboard.

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Definition (1.)3.2

The set with no elements is called the **empty set** and is denoted $\emptyset$.

The Axioms

The axiom schema of comprehension

Let $\mathbf{P}(x)$ be a unary property of x . For any set A , there is a set B such that $x \in B$ if and only if $x \in A$ and $\mathbf{P}(x)$.

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The above axioms is a **schema** of axioms, that is, there is an axiom for each property $\mathbf{P}(x)$.

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Example

Let $\mathbf{P}(x)$: $x \notin x$. The axiom postulates:

For any set A , there is a set B such that $x \in B$ if and only if $x \in A$ and $x \notin x$ (the set B is the empty set).

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$$\forall A \exists B \forall x (x \in B \leftrightarrow x \in A \wedge \mathbf{P}(x)).$$

Remark

“Although the supply of axioms is unlimited, this causes no problems, since it is easy to recognize whether a particular statement is or is not an axiom and since every proof uses only finitely many axioms.” (p. 8)

The Axioms

Lemma

If P and Q are sets, then there is a set R such that $x \in R$ if and only if $x \in P$ and $x \in Q$.

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Proof

Hint: To define the property $\mathbf{P}(x, Q): x \in Q$ and use the axiom schema of comprehension.

References



Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on p. 2).